



TECHNOLOGY
SOLUTIONS UK LTD
part of **HID**



Comparison of *Bluetooth*[®] modes for TSL Readers

OVERVIEW

TSL 1st and 2nd Generation UHF RFID Readers

TSL 11xx/21xx readers (e.g. 1128, 2128P, 2166) support three modes of operation:

- **USB connection** - using TSL ASCII 2 Protocol
- **Bluetooth SPP Mode** - a Bluetooth classic connection using ASCII 2 Protocol
- **Bluetooth HID Mode** - a Bluetooth classic connection using the HID profile (the TSL reader appears as a Bluetooth keyboard)

TSL 3rd Generation UHF RFID Readers

TSL 31xx readers (e.g. 3138 and 3166) support five modes of operation:

- **USB connection** - using TSL ASCII 3 Protocol (backwardly compatible with ASCII 2)
- **Bluetooth SPP Mode** - a Bluetooth classic connection using ASCII 3 Protocol
- **Bluetooth HID Mode** - a Bluetooth classic connection using the HID profile (the TSL reader appears as a Bluetooth keyboard)
- **Bluetooth Serial Over GATT (SOG)** - the Bluetooth Low Energy equivalent of SPP mode, also using ASCII 3 Protocol
- **Bluetooth HID Over GATT (HOG)** - the Bluetooth Low Energy equivalent of HID mode (the TSL reader appears as a Bluetooth keyboard)

Instructions for changing Bluetooth modes can be found in Appendix A

The comparison chart below highlights the differences between serial and HID operating modes:

BLUETOOTH MODES COMPARISON

Serial Modes	HID Modes
• Bluetooth SPP • Bluetooth Low Energy SOG	• Bluetooth HID • Bluetooth Low Energy HOG
Apps need to be written with specific support for the Reader	Apps can use the reader without modification
The reader can be discovered and paired in the Bluetooth Settings or from within the App but the App controls the connection • Connects when instructed by the App • Disconnects when instructed by the App • The reader powers off when no longer connected and idle • The App needs to connect to restore	Reader is discovered and paired in the Host Bluetooth Settings (often appears as a keyboard) • After pairing the reader connects automatically • If idle the reader sleeps and the connection is dropped • The reader, when woken, automatically reconnects to the host device
Apps receive and interpret ASCII Protocol responses when the user triggers a UHF or barcode scan	Apps receive input as Key strokes from the reader including Tab and Return/Enter keys

COMPARISON (CONTINUED)

Serial Modes • Bluetooth SPP • Bluetooth Low Energy S0G	HID Modes • Bluetooth HID • Bluetooth Low Energy H0G				
The App can respond and react intelligently to responses e.g. • duplicate responses can be ignored or counted • incoming tag data can be truncated, stripped or transformed into alternative representations: Hex, ASCII, GS1 EPC URI, etc...	The reader types text for each barcode/UHF scan received. • The tag values can be returned in Hex or ASCII representations <table border="1" data-bbox="810 633 1476 958"> <thead> <tr> <th data-bbox="810 633 1141 689">TSL 11xx/21xx Readers</th><th data-bbox="1147 633 1476 689">TSL 31xx Readers</th></tr> </thead> <tbody> <tr> <td data-bbox="810 698 1141 958"> • Up to 2 Additional characters can be inserted before and after the text sent • The reader cannot truncate or strip values from tag data </td><td data-bbox="1147 698 1476 958"> • Extra text may be included in the data returned • The reader can trim tag/barcode data </td></tr> </tbody> </table>	TSL 11xx/21xx Readers	TSL 31xx Readers	• Up to 2 Additional characters can be inserted before and after the text sent • The reader cannot truncate or strip values from tag data	• Extra text may be included in the data returned • The reader can trim tag/barcode data
TSL 11xx/21xx Readers	TSL 31xx Readers				
• Up to 2 Additional characters can be inserted before and after the text sent • The reader cannot truncate or strip values from tag data	• Extra text may be included in the data returned • The reader can trim tag/barcode data				
The App can change the Reader's behaviour	The Reader can only send scanned data to the host, the host cannot change the reader's behaviour				
The Reader's behaviour and command parameters are controlled in real time by the App e.g. the trigger action can change to suit the task that the User is performing; the App can allow the User to specify Inventory output power	The Reader's behaviour, such as the action of the reader's trigger switch, the inventory output power, the idle sleep timeout and other command parameters can be configured only once at start-up.				
Configuration is held within the App (any configuration in the AUTO.TXT file is likely to be overridden by the App settings)	All configuration is held in an AUTO.TXT file stored on an SD-Card, or in internal storage. Removing the SD-card or deleting the AUTO.TXT and power-cycling the unit restores default settings.				
All reader activity, by default, is saved to a log file if an SD-Card is fitted	All reader activity, by default, is saved to a log file if an SD-card is fitted				

APPENDIX A

Introduction:

TSL Bluetooth Readers can be operated in *SPP mode* where the Reader is controlled by a custom-written application or in *HID mode*, where the Reader behaves like a Bluetooth Keyboard.

The HID mode has two variants¹ :

1. *HID mode* Compatible with Android, Windows and Apple devices
2. *HID Apple mode* Compatible only with Apple devices providing an on-screen keyboard toggle

The Bluetooth operating mode of a TSL Reader can be changed using the steps detailed below.

CHANGING THE BLUETOOTH MODE USING TSL SOFTWARE

Preparation:

Download and install 'TSL PC Firmware Downloader' from the product's [Downloads Page](#) (free, one time, registration required)

To switch to Bluetooth HID mode:

- First delete existing pairings to the TSL Reader
- Launch the downloader (*Desktop Firmware Loader*)
- Connect the TSL Reader to the computer using the supplied USB cable.
- The driver should install automatically (*)
- The downloader should prompt as the com port arrives and select the com port automatically ("COMn arrived")
- Select **Action>Reset Bluetooth to HID mode** or **Action>Reset Bluetooth to HID Apple mode** (wait for the process to complete)
- Disconnect the TSL Reader from the computer (unplug USB)
- Wake the TSL Reader (pull trigger, blue flashing LED)
- Pair to the reader on the device (it typically appears as a keyboard icon and connects automatically)
- Scanned data should now be typed into the current application on the device

To switch to Bluetooth SPP mode:

- First delete existing pairings to the TSL Reader
- Launch the downloader (*Desktop Firmware Loader*)
- Connect the TSL Reader to the computer using the supplied USB cable.
- The driver should install automatically (*)
- The downloader should prompt as the com port arrives and select the com port automatically ("COMn arrived")

- Select **Action>Reset Bluetooth to SPP mode** (wait for the process to complete)
- Disconnect the TSL Reader from the computer (unplug USB)
- Wake the TSL Reader (pull trigger, blue flashing LED)
- Pair to the reader on the device
- Launch an Application (e.g. *RFID Explorer*) to use the reader

(*) if not download from here <https://ftdichip.com/drivers/vcp-drivers/>

Further information can be found from the Reader download pages (free, one time, registration required - see links above)

[Application Note – Bluetooth HID mode](#): Provides further examples for configuring HID mode

[Application Note – Using the Micro SD log and auto-run files](#): Describes using the AUTO.TXT

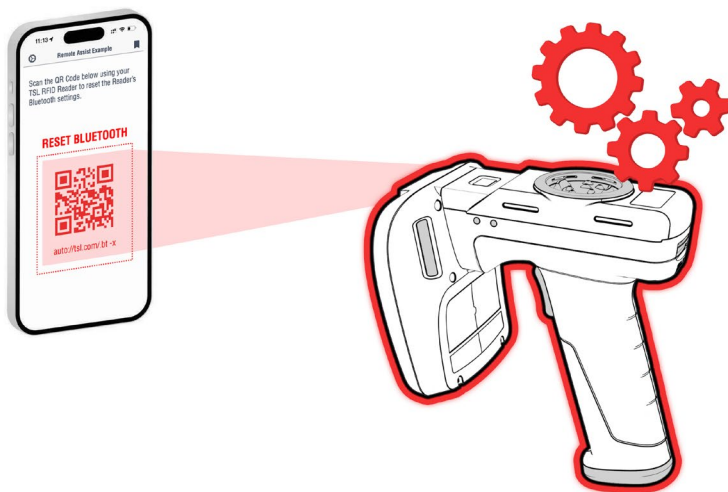
[TSL ASCII Protocol 2.5 Rev A](#): Details all the commands including for HID .st, .hs, .hc, and .hd

If you encounter any issue while following the above procedure please contact support@tsl.com providing the following information:

- The serial number of the TSL Reader
- The Firmware version of the TSL Reader
- The step at which the procedure failed
- A description of the fault that occurred

Firmware requirements: 1128 - v4.2.7 or greater; 1153 – v2.1.7 or greater; 1166 – v1.1.6 or greater

CHANGING THE BLUETOOTH MODE USING COMMAND SCAN



TSL 31xx readers equipped with a 2D barcode imager can make use of [Command Scan](#) to configure the reader by scanning special 2D barcodes. Further information can be found from the following document:

[Command Scan Examples](#): Contains examples of Command Scan configuration barcodes, including changing Bluetooth Mode by simply scanning a 2D barcode. Links to video examples are also provided, along with a breakdown of the ASCII commands contained within each example.

About

ABOUT TSL



Technology Solutions UK Ltd (TSL®), part of HID, is a leading manufacturer of high performance mobile RFID readers used to identify and track products, assets, data or personnel.

For over two decades, TSL has delivered innovative data capture solutions to Fortune 500 companies around the world using a global network of distributors and system integrators. Specialist in-house teams design all aspects of the finished products and software ecosystems, including electronics, firmware, application development tools, RF design and injection mould tooling.

TSL is an ISO 9001:2015 certified company.



ISO 9001: 2015

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ABOUT HID



HID powers the trusted identities of the world's people, places and things. We make it possible for people to transact safely, work productively and travel freely. Our trusted identity solutions give **people** convenient access to physical and digital **places** and connect **things** that can be identified, verified and tracked digitally. Millions of people around the world use HID products and services to navigate their everyday lives, and billions of things are connected through HID technology. We work with governments, educational institutions, hospitals, financial institutions, industrial businesses and some of the most innovative companies on the planet. Headquartered in Austin, Texas, HID has over 4,000 employees worldwide and operates international offices that support more than 100 countries. HID is an ASSA ABLOY Group brand.

For more information, visit www.hidglobal.com.